

- Outline how a frame quadrat could be used to assess the biodiversity of plants in a field.

(b) A student investigated whether the height of the soft rush plant, *Juncus effusus*, decreases with an increase in altitude on a hillside in the United Kingdom.

- Spearman's rank correlation was used to assess the relationship between the height of the plants and altitude.

$$r_s = 1 - \left(\frac{6 \times \Sigma D^2}{n^3 - n} \right)$$

Key to symbols:
 D = difference in rank between each pair of measurements
 n = number of pairs of items in the sample

(i) Calculate the Spearman's rank correlation for these data.

answer [3]

- (ii) The null hypothesis for this investigation is: there is no correlation between the altitude and the height of the soft rush plants.

Table 9.1 shows the critical values for Spearman's rank correlation.

Table 9.1

<i>n</i>	<i>p</i> = 0.05
10	0.564
12	0.504
14	0.459

Use your value of Spearman's rank correlation and Table 9.1 to state **and** explain if the null hypothesis is correct.

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..... [2]

[Total: 9]